

TechnologyStack

Date	14 July2026
TeamID	SWUID2026221635
ProjectName	A Comprehensive Measure of Well-Being (HDI Prediction & Analysis)
Maximum Marks	2 Marks

DefineYourTechnologyStack:

Identifytheprogramminglanguages,frameworks,databases,front-endtools,back-endtools,and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Filloutthetablebelowtodetailthespecifictechnologieschosenforeachcomponentofyour architecture and provide a brief justification for your choice.

TechnologyStackDetails

S.No	Architecture Component/Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g.,React,Angular, HTML/CSS)</i>	HTML5, CSS3, JavaScript (with Tailwind CSS / Bootstrap)	Highly intuitive, easy to build responsive forms, and displays dashboard charts perfectly on all screen sizes.

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django,Java)</i>	Python (using Flask Framework)	Flask is lightweight, highly efficient for web application routing, and connects seamlessly with Python's Machine Learning libraries.
3	Database/Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	MongoDB	MongoDB is lightweight, serverless, and perfect for managing local user records and prediction histories efficiently.
4	Cloud/Hosting/ Deployment <i>(e.g.,AWS,Azure, Heroku, Docker)</i>	Render or Heroku	Both provide easy, cost-effective, and fast cloud hosting platforms with direct GitHub deployment integrations.
5	VersionControl& CI/CD	Git and GitHub	Essential for collaborative team tracking, code versioning, and hosting the project portfolio files.

	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs/ Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	Scikit-Learn, Pandas, NumPy, Matplotlib	Pandas and NumPy handle data preprocessing, Scikit-learn trains the predictive ML model, and Matplotlib renders the analytical charts.